

TIRTHARAJ BHATTACHARYA

Electronics & Communication Engineering Student | GNIT, Kolkata

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[Linkedin](#) | [Github](#) | [Portfolio](#)



About Me

B.Tech (ECE) undergraduate with interests in Embedded Systems, Artificial Intelligence, IoT, and software development, with a focus on integrating intelligent behaviour into real-world systems. Worked on collaborative projects involving embedded systems, web applications, and exploratory AI/ML implementations, gaining practical exposure to machine learning workflows and embedded technologies. Interested in building context-aware and adaptive systems through the intersection of hardware, software, and AI.

Education

Guru Nanak Institute of Technology – B.Tech, ECE (2024–Present)

Central Model School, Barrackpore – AISSCE, PCMB (2021–2023): 90%

Central Model School, Barrackpore – AISSE (2008–2021): 94%

DAV Public School, Barrackpore – (2008–2019)

Experience

AIML Trainee, Ardent Computech Pvt Ltd | Internship (March, 2026 – April, 2026)

Content Writer, Euphonous X Zyzzva 2k26 | Volunteer (December, 2025 – March, 2026)

Content Writer, Euphonous X Zyzzva 2k25 | Volunteer (February, 2025 – September 2026)

Technical Skills

Programming Languages: Python, C, HTML, CSS

AI/ML & Libraries: TensorFlow, Keras, OpenCV, NumPy, Pandas

Hardware & Tools: Arduino Uno, ESP32, IoT Fundamentals, Breadboarding

Core Competencies:

- Public Speaking (Debates & Speeches)
- Leadership
- Team Collaboration
- Content Writing
- Presentations

Research and Publications

Research Paper – Co-author

Design And Development Of A Resonance-Tuned Hybrid Piezoelectric-Electromagnetic Energy Harvesting System (*Published in IJCRT*)

- Contributed to manuscript editing, refinement, and review alongside co-authors.
- Participated in collaborative discussions and content preparation during the publication process.

Projects

NeuroKavach | Idea Manthan Project (2nd Position)

Technologies: Haptic Feedback Systems, Wearable Technology

- Proposed a wearable haptic feedback concept for real-time cognitive intervention
- Collaborated with teammates to refine functionality and presentation
- Secured 2nd position through concept development and solution presentation

Digital Fort Knox | Cryptonix-V1 Project

Technologies: ESP32, IoT, Device Monitoring

- Worked with a team to develop a security-focused embedded solution
- Contributed to solution design, feature discussions, and prototype development.

Emotion Detection Project | BCT Training Program

Technologies: Python, OpenCV

- Implemented an emotion detection system during technical training activities.
- Gained exposure to computer vision workflows and external library integration.
- Learned practical implementation alongside theoretical Python concepts.

MatsyAI | Hackforge Project

Technologies: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn

- Worked collaboratively in a team to design and develop a proposed solution
- Participated in brainstorming, solution planning, and project development within a time-constrained environment

Plant Disease Detector

Technologies: Python, TensorFlow

- Built an experimental machine learning project for identifying plant diseases from image data.
- Explored image preprocessing and basic classification workflows.
- Used the project as a practical introduction to machine learning concepts.

Screen Recorder

Technologies: Python, Pyautogui

- Developed a utility for capturing on-screen activity using Python libraries.
- Explored system interaction and screen capture functionality.

Voice Recorder

Technologies: Python, NumPy, SoundDevice

- Developed a utility for recording audio input using Python audio-processing libraries.
- Explored concepts related to audio capture and handling.

Kitab-Al Hind (Book Selling Website)

Technologies: HTML, CSS

- Built a responsive website interface for displaying and organizing book information.

Achievements and Competitions

- Earned 2nd Position at *Idea Manthan* organized by IEM for **NeuroKavach**
- Earned 1st Position at *Eureka* organized by GNIT for **Gurdian Mat**, contributing to concept presentation.
- Participated in *Cryptonix-VI* organized by Heritage Institute of Technology with **Digital Fort Knox**
- Selected at Institute-level screening for **Smart India Hackathon (SIH)**
- Participated in *Hackforge* organized by Jadavpur University with **MatsyAI**

Positions of Responsibility

Vice President – GNIT Literary Club | 2025 – Present

Content & Communication Activities

- **Content Creator — The Anime Maniacs (YouTube)**
Built a content channel reaching **250K+ cumulative views** and **400+ subscribers**
- **Freelance Blogger**
Published **10+ articles** across online platforms
- Participated in multiple technical competitions and collaborative events
- Participated in several debates, and other literary activities

Certifications

[Programming for Everybody \(Getting Started with Python\)](#) — University of Michigan

[Python Data Structures](#) — University of Michigan

[AI Tools and ChatGPT Workshop](#) — Be10x

Languages

English — Fluent

Bengali — Native

Hindi — Conversational